

Introduction

We learned a little about equivalence relations several weeks ago, but now it's time to put that knowledge to work. Our knowledge of divisibility will afford us the chance to define a very important type of equivalence relation.

Learning Objectives

- Define congruence modulo n and be able to show that it is an equivalence relation.
- Show that addition and multiplication are well-defined in modular arithmetic.
- Define $\mathbb{Z}_n$, and describe how the properties of $\mathbb{Z}_n$ change depending on whether n is prime.
- State and prove Fermat's Little Theorem.

Modular Equivalence

Recall that a relation $\sim$ on a set A is said to be an *equivalence relation* if R is reflexive, transitive, and symmetric. In particular, $\sim$ satisfies

- *reflexive* if for all $a \in A$, $a \sim a$.
- *symmetric* if for all $a, b \in A$, $a \sim b$ implies that $b \sim a$.
- *transitive* if for all $a, b, c \in A$, $a \sim b$ and $b \sim c$ implies that $a \sim c$.

Definition 1

If $n \in \mathbb{Z}^+$ and $a, b \in \mathbb{Z}$, we say that $a \equiv b \pmod{n}$ (read: a is congruent to $b \pmod{n}$) if $n \mid (b - a)$.

For example, if $n = 4$ then $1 \equiv 29 \pmod{4}$, since $4 \mid (29 - 1) = 28$.

Proposition 2

Congruence mod n is an equivalence relation.

Proof. We need to show that congruence is reflexive, transitive, and symmetric.

- [Reflexive] Fix an $a \in \mathbb{Z}$. Note that $a - a = 0$ and $n \mid 0$, so $a \equiv a \pmod{n}$.
- [Transitive] Suppose that $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, or equivalently $n \mid (b - a)$ and $n \mid (c - b)$. Now

$$(c - a) = (c - b + b - a) = (c - b) + (b - a)$$

and since n divides each of above terms, we know it divides the sum as well. Thus $n \mid (c - a)$, or $a \equiv c \pmod{n}$.

- [Symmetric] Suppose that $a \equiv b \pmod{n}$, so that $n \mid (b - a)$. Then $n \mid (a - b) = -(b - a)$, so $b \equiv a \pmod{n}$ as required. $\square$

More importantly, we can add and multiply congruence relations.

Proposition 3

Fix an $n \in \mathbb{Z}^+$. If $a \equiv r \pmod{n}$ and $b \equiv s \pmod{n}$ then

1. $(a + b) \equiv (r + s) \pmod{n}$.
2. $ab \equiv rs \pmod{n}$

Proof. Fix a, b, r, s and assume that $a \equiv r \pmod{n}$ and $b \equiv s \pmod{n}$. Hence $n \mid (r - a)$ and $n \mid (s - b)$. Equivalently, there exist k, ℓ such that $nk = r - a$ and $n\ell = s - b$.

1. Summing together our two equations above, we get

$$n(k + \ell) = (r + s) - (a + b)$$

so $n \mid [(r + s) - (a + b)]$ or equivalently, $(a + b) \equiv (r + s) \pmod{n}$

2. We can write

$$rs - ab = rs - rb + rb - ab = r(s - b) + (r - a)b = n\ell r + nkb = n(\ell r + kb)$$

so $n \mid (rs - ab)$ or equivalently, $ab \equiv rs \pmod{n}$. $\square$

For a fixed $n \in \mathbb{Z}^+$ we define the *congruence class* of $a \in \mathbb{Z}$ to be the equivalence class

$$[a] = \{x \in \mathbb{Z} : x \equiv a \pmod{n}\},$$

of which there are precisely n ; namely $[0], [1], [2], \dots, [n - 1]$. We often denote this set of equivalence classes by

$$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z} = \{[0]_n, [1]_n, \dots, [n - 1]_n\}.$$

According to Proposition 3, we can add and multiply congruence classes exactly as we would integers, as long as we reduce modulo n . So for example, working modulo 4 we have

$$[3]_4 + [2]_4 = [5]_4 = [1]_4, \quad [3]_4 \cdot [2]_4 = [6]_4 = [2]_4.$$

Indeed, we can create the addition and multiplication tables for $\mathbb{Z}_4$ as follows:

+	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$\times$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$
$[1]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[2]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[0]_4$	$[2]_4$	$[0]_4$	$[2]_4$
$[3]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[3]_4$	$[2]_4$	$[1]_4$

It is interesting to note that some, but not all, of the elements of $\mathbb{Z}_4$ have multiplicative inverses. For example, $[3]_4 \cdot [3]_4 = [1]_4$. We will see later that if our modulus is prime, then in fact all non-zero equivalence classes have a multiplicative inverse!

Example 4

What is the last digit in 4^{441} ?

Solution. Note that the last digit d satisfies $d \equiv 4^{441} \pmod{10}$. Moreover,

$$4^{441} = (4^3)^{147} = (64)^{147} \equiv 4^{147} \pmod{10}.$$

The same trick again gives

$$4^{147} = 64^{49} \equiv 4^{49} \pmod{10}.$$

Here we need to be a bit more clever. Rather than try to decompose this into $49 = 7 \cdot 7$, let's write $49 = 1 + 48$ so that

$$\begin{aligned} 4^{49} &= 4 \cdot 4^{48} = 4 \cdot (64)^{16} \equiv 4 \cdot 4^{16} \pmod{10} \\ &\equiv 4 \cdot (256)^4 \pmod{10} \equiv 4 \cdot 6^4 \pmod{10} \\ &\equiv 4 \cdot (36)^2 \pmod{10} \equiv 4 \cdot 6^2 \pmod{10} \\ &\equiv 4 \cdot 36 \pmod{10} \equiv 4 \cdot 6 \pmod{10} \\ &\equiv 4 \pmod{10}. \end{aligned}$$

Hence the last digit of 4^{441} is 4. ■

Exercise 5 Show that if $a^2 - 2$ is not divisible by 7, then $a - 4$ is not divisible by 7.

The Integers Mod p

We have learned that we can add and multiply the elements of $\mathbb{Z}_n$ similar to usual integers. However, when the modulus is *prime*, it turns out we can say even more! To start, let's introduce a small but useful lemma.

Lemma 6

Suppose p and x are integers such that $\gcd(p, x) = 1$. If $m, n \in \mathbb{Z}$ and $mx \equiv nx \pmod{p}$ then $m \equiv n \pmod{p}$.

Proof. The statement that $mx \equiv nx \pmod{p}$ implies that $p \mid (mx - nx)$, or equivalently that $p \mid (m - n)x$. Since p and x are coprime, this means that $p \mid (m - n)$; that is, $m \equiv n \pmod{p}$. □

This lemma tells us that we can “cancel” the x ’s in the equation $mx \equiv nx \pmod{p}$ so long as p and x are coprime. To see that this cancellation does not hold in general, note that

$$3 \cdot 2 \equiv 1 \cdot 2 \pmod{4}; \quad \text{however, } 3 \not\equiv 1 \pmod{4}.$$

Hence the condition that p and x be coprime is necessary. In our context, we will often take p to be prime, meaning that so long as x is not a multiple of p ($x \not\equiv 0 \pmod{p}$) then we will be able to cancel to our heart’s content.

Theorem 7: Fermat’s Little Theorem

If $a \in \mathbb{Z}, p \in \mathbb{N}$ are such that p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the numbers $S = \{0a, 1a, 2a, \dots, (p-1)a\}$. We claim that no two of these elements lie in the same equivalence class modulo p . Indeed, since $p \nmid a$ then $\gcd(p, a) = 1$. Thus if $ma \equiv na \pmod{p}$ then $m \equiv n \pmod{p}$, showing that the elements of S all lie in different equivalence classes. Since S consists of exactly p elements and so too does $\mathbb{Z}_p$, the elements of S are just a rearrangement of the elements of $\mathbb{Z}_p$.

Multiplying all of the non-zero elements of S together will thus give the same result as multiplying all of the non-zero elements of $\mathbb{Z}_p$ together, so

$$\begin{aligned} (a)(2a) \cdots ((p-1)a) &\equiv (1)(2) \cdots (p-1) \pmod{p} \\ (p-1)!a^{p-1} &\equiv (p-1)! \pmod{p}. \end{aligned}$$

Once again, we see that $\gcd(p, (p-1)!) = 1$ (why?) and so

$$a^{p-1} \equiv 1 \pmod{p},$$

as required. □

A straightforward corollary of this is the following, which again does not hold if p is not prime (why?).

Corollary 8

For any integer a and prime p , $a^p \equiv a \pmod{p}$.

Exercise 9 Prove Corollary 8, noting that the restriction that $p \nmid a$ from Fermat’s Little Theorem is not present.

All of this together allows us to make an interesting statement about the existence of multiplicative inverses in $\mathbb{Z}_n$. Namely, if n is a prime number, then every non-zero element of $\mathbb{Z}_n$ has a multiplicative inverse

Theorem 10

If p is prime, then every non-zero element of $\mathbb{Z}_p$ has a multiplicative inverse.

Proof. Suppose that p is a prime and let $a \in \mathbb{Z}_p$ be a non-zero element. Since p is prime and $a \neq 0$ we have $p \nmid a$, so by Fermat's Little Theorem we know that

$$a \cdot a^{p-2} = a^{p-1} \equiv 1 \pmod{p},$$

thus $a^{-1} = a^{p-2}$ is the multiplicative inverse to a . $\square$

This proof doesn't require Fermat's Little Theorem, and there is a much easier proof! Can you think of it? It will help you significantly for the next exercise.

Exercise 11 Find the inverse of $[22]$ in $\mathbb{Z}_{41}$.

Exercises

- Determine whether exponentiation is well-defined in modular arithmetic. In particular, suppose that $n \in \mathbb{Z}^+$ with $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Determine whether $a^c \equiv b^d \pmod{n}$.
- Find the last (one's) digit of the following numbers:
 - 7^{100}
 - 3^{1111}
 - 4^{223}
 - 8^{400}
- The *root* of a number is the single digit number formed by recursively summing its digits. So for example,

$$15993 \mapsto 1 + 5 + 9 + 9 + 3 = 27 \mapsto 2 + 7 = 9.$$

So the root of 15993 is 9. Let $r(n)$ denote the root of n . Show that $r(n) = 1 + [(n-1) \pmod{9}]$.

- Show that $9 \mid (10^n - 1)$ for all $n \in \mathbb{Z}^+$.
- Show that for any $n \in \mathbb{Z}^+$, n^2 is equivalent to either 0 or 1 mod 4.
- Determine if $7 \mid 3^{2001} + 4^{2001}$.
- Suppose that p is prime.
 - Show that if $a^2 \equiv b^2 \pmod{p}$ then $a \equiv \pm b \pmod{p}$.
 - Show that the above result is not true if p is not prime.